A binary cubic 3-gluing recognition criterion

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Status: derived criterion and proof; publication priority unresolved. This is a different search direction from the existing hyperelliptic Artin–Schreier sweeper. It does not establish a novel affected family or a completed attack. Beating Pollard rho is not an acceptance condition.

Requested deliverable	Current evidence
A new affected family or recognition criterion	Candidate derived; publication novelty unresolved.
A short proof	Three derived statements below; no Lean or independent mathematical review.
Rust in the cryptanalysis repository	Standalone native checker, with independent finite-algebra unit verification.
An explicit subgroup/log transfer	Not implemented; target, torsion coordinates and twist character open.
Continue the earlier search	355 hyperelliptic jobs still pending; protocol admission unresolved.

Question and literature boundary

Can a small-torsion test identify binary elliptic curves for which a genus-three transfer exists, without first enumerating hyperelliptic covers? Tian's 2020 preprint, section 3, already constructs cubic-extension transfers through rational maximal isotropic subgroups of a Weil restriction. The approach itself is prior work. The preprint's reconstruction uses odd-characteristic equations and two-torsion bitangents; it is not a binary implementation. The 2023 journal version is subscription-only in this session; its full contents have not been read.

The narrower contribution attempted here is a complete mod-3 module test, explicit kernels, and a binary sufficient condition for geometric indecomposability. Novelty of that formulation still needs a direct check against the full journal version and other gluing classifications. Unsuccessful keyword searches do not settle priority.

An exact module criterion

Let $k = \mathbf{F}_q$, $K = \mathbf{F}_{q^3}$, with characteristic different from three, and let E/K be elliptic. Put $A = \operatorname{Res}_{K/k} E$. Let M be the matrix of q^3-Frobenius on $E[3]$, so $\det M = q^3 = q \pmod 3$, and let $t = \operatorname{tr} M$ in $\mathbf{F}_3$. In Frobenius-transported bases, $A[3] = V_0 \oplus V_1 \oplus V_2$ has

$$F(v_0, v_1, v_2) = (Mv_2, v_0, v_1), \qquad \Omega = J \oplus qJ \oplus q^2 J, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Thus $F^*\Omega = q\Omega$. Call a three-dimensional subspace L **coordinate-transverse** if $L \cap V_i = 0$ for each i. This is the necessary coordinate condition in Tian's construction, not by itself a general theorem that a quotient is a Jacobian. Here a rational subgroup means a subgroup scheme defined over k; its generators may have coordinates in an extension.

Proposition (derived). An F-stable, coordinate-transverse Lagrangian $L \subset A[3]$ exists if and only if either

1. $q = -1 \pmod 3$ and $t = 0 \pmod 3$; or
2. $q = 1 \pmod 3$ and $M = I$ or $M = -I$.

Positive cases. In case 1 choose eigenvectors e, f of M with eigenvalues $1, -1$. Write their transported copies as e_i, f_i. The subspace

$$L = \langle e_0 + e_1 + e_2,\ f_0 + f_1,\ f_1 + f_2 \rangle$$

is stable: F cyclically permutes the e_i, sends f_0 to f_1, f_1 to f_2, and f_2 to $-f_0$. The f-coordinates satisfy $b_0 - b_1 + b_2 = 0$. The weighted pairing makes them orthogonal to $e_0 + e_1 + e_2$. There are three independent generators, so L is Lagrangian. A vector supported in one factor must first have zero e-coefficient, then zero f-coefficient by that relation.

In case 2 write $M = \lambda I$, $\lambda = \pm 1$. Rescaling the bases of the three factors by $1, \lambda, 1$ turns F into λ times cyclic permutation and leaves the pairing unchanged. In these bases put

$$L = \{((s, y_0), (s, y_1), (s, y_2)) : s = y_0 + y_1 + y_2\}.$$

It has dimension three, is stable, and its pairing on two vectors is $s \sum y_i' - s' \sum y_i = 0$. Support in one factor forces $s = 0$ and then the remaining $y_i = 0$. Transform back to the transported bases.

Exclusion of all other cases. The characteristic polynomial of F is

$$T^6 - tT^3 + q^3 = (T^2 - tT + q)^3 \quad \text{over } \mathbf{F}_3.$$

When $q = -1, t \neq 0$, or $q = 1, t = 0$, the quadratic on the right is irreducible. Every invariant subspace then has even dimension, excluding a Lagrangian of dimension three. In the remaining case $q = 1, t \neq 0$, M has a repeated eigenvalue $\lambda = \pm 1$. If it is nonscalar, $\dim \ker(F - \lambda) = \dim \ker(M - \lambda) = 1$, so F is a single Jordan block of size six. Its unique invariant three-space is $\ker(F - \lambda)^3 = \ker(M - \lambda)^{\oplus 3}$. It intersects each coordinate plane in a line. This proves necessity. □

A binary coefficient test

For the ordinary normalized binary curve

$$E/K : \quad y^2 + xy = x^3 + ax^2 + b, \qquad b \neq 0,$$

the three-division polynomial is $\psi_3(X) = X^4 + X^3 + b$. Its four distinct roots encode the four lines in $E[3]$; they depend on b, not on the quadratic twist a. They are distinct because $\psi_3' = X^2$ and $b \neq 0$. The proposition therefore becomes the following exact recognition rule:

Base field	Rational roots of ψ_3 over K required
$q = 2^d$, d odd	Exactly two, equivalently factor degrees $(1, 1, 2)$.
$q = 2^d$, d even	All four, equivalently factor degrees $(1, 1, 1, 1)$.

For odd d, determinant -1 makes two fixed projective lines equivalent to eigenvalues $1, -1$; the alternative is an irreducible quadratic. For even d, all four lines are fixed exactly when M is scalar. Root count is $\deg \gcd(\psi_3, X^{2^{3d}} - X)$, requiring no elliptic point-counting census. For odd d one may also use $3 \mid \#E(K)$, since $q^3 + 1 = 0 \pmod 3$ and $t = 0 \pmod 3$.

The odd-degree rule reduces to one trace

When d is odd, set $n = 3d$ and $c = b^{(2^{n+1}-1)/3}$, the unique cube root of b in K. The coefficient test is exactly

$$\mathrm{Tr}_{K/\mathbf{F}_2}(c) = 0.$$

Here is a direct algebraic proof. Cubing is bijective for odd n. The map $u \mapsto u^4 + u$ has kernel $\mathbf{F}_2$ and image the absolute-trace-zero hyperplane. The substitutions

$$c = u^4 + u = u(u^3 + 1), \qquad x = u^3 + 1$$

give $x^4 + x^3 = c^3 = b$. Conversely, from a root x take $u = \sqrt[3]{x+1}$ to recover $c = u^4 + u$. Thus the quartic has roots if and only if c has trace zero, and then has exactly two. An explicit solution of $u^4 + u = c$ is the Frobenius-four half-trace

$$u = \sum_{i=0}^{(n-1)/2} c^{2^{4i}}.$$

Indeed $u^4 + u = c + \mathrm{Tr}(c)$. Its two solutions $u, u+1$ give the two division roots $u^3 + 1, (u+1)^3 + 1$.

This trace test is not new on its own. Its connection to rational three-torsion and binary Kloosterman sums overlaps established work; see Moisio, Theorem 7.3 and Remark 7.4, which cites Charpin–Helleseth–Zinoviev for the odd-degree case. That source was checked before adding this simplification. The candidate under priority review is its use with the induced-module and indecomposability arguments here, not a claim of inventing the trace condition.

From the kernel to a genus-three Jacobian

Lemma (derived). Suppose no degree-one or degree-two isogeny over $\bar{k}$ joins two distinct factors of $A_{\bar{k}}$. If L is as above, the induced principally polarized quotient $B = A/L$ is geometrically indecomposable.

Proof. The quotient isogeny $\phi : A \to B$ has polarized dual $\psi : B \to A$ with $\psi\phi = [3]$, $\phi\psi = [3]$, and $\psi^*\lambda_A = 3\lambda_B$. If B were a product, its dimension three forces an elliptic polarized factor $\iota : D \hookrightarrow B$. The components f_i of $\psi\iota : D \to \prod E_i$ satisfy $\sum_i \deg f_i = 3$, with zero assigned to a zero component. If at least two are nonzero, one has degree one, and another has degree one or two. Their composition gives a forbidden isogeny between distinct E_i. If just one component is nonzero it has degree three. Since $\phi\psi\iota = [3]\iota$, the corresponding $\phi|_{E_i} : E_i \to D$ also has degree three. Its kernel is a nontrivial subgroup of $L \cap E_i[3]$, contradicting transversality. □

Corollary (derived; one twist only). Let $q = 2^d$, $d \geq 2$, and let $b \in \mathbf{F}_{2^{3d}}$ have full **absolute** degree $3d$. If the root criterion above holds, one of the two quadratic twists of E admits a k-rational $(3,3,3)$-isogeny

$$\mathrm{Res}_{K/k} E^{(\epsilon)} \longrightarrow J_C,$$

where C/k has genus three. The isogeny is injective on every prime-order subgroup with prime different from three.

Proof. The conjugate j-invariants are $1/b$, $1/b^q$, $1/b^{q^2}$. Full absolute degree makes them distinct. For an ordinary binary elliptic curve, every geometric degree-two isogeny has target j^2 or $\sqrt{j}$: its kernel is the unique connected or the unique etale order-two subgroup. The exponents $d, 2d$ modulo $3d$ are neither 1 nor -1 when $d \geq 2$. Thus the lemma applies to the three conjugate factors.

The quotient is a geometrically indecomposable principally polarized threefold. The dimension-three Jacobian theorem and arithmetic Torelli, as stated in Beauville–Ritzenthaler, section 3, give $B \simeq J_C$ or its $[-1]$ quadratic twist. These existence statements include characteristic two; their odd-characteristic discriminant

reconstruction is not used here. Since $[K : k] = 3$ is odd, the nontrivial quadratic character of k restricts nontrivially to K. Twisting the isogeny therefore selects one of the two source twists. Its kernel is killed by three, so a prime subgroup of order different from three survives. □

In particular, for every odd $d \geq 3$ this gives the succinct sufficient criterion: full absolute degree of b and $\mathrm{Tr}(\sqrt[3]{b}) = 0$ imply a genus-three transfer for one source twist. A parametric version is $b = (u^4 + u)^3$, retaining only nonzero full-degree b. This is a derived family description; its status as a previously unrecognized affected family has not been established.

This proves an existence criterion for a twist pair. It does **not** identify the right source twist, instantiate C, evaluate the isogeny, or prove that the resulting C is non-hyperelliptic. At $d = 1$ the degree-two conjugate isogenies invalidate the lemma's hypothesis. Proper-subfield and Koblitz coefficients are excluded from the corollary. Failure of the criterion excludes this particular 3-gluing kernel type; other isogeny degrees and cover constructions remain possible.

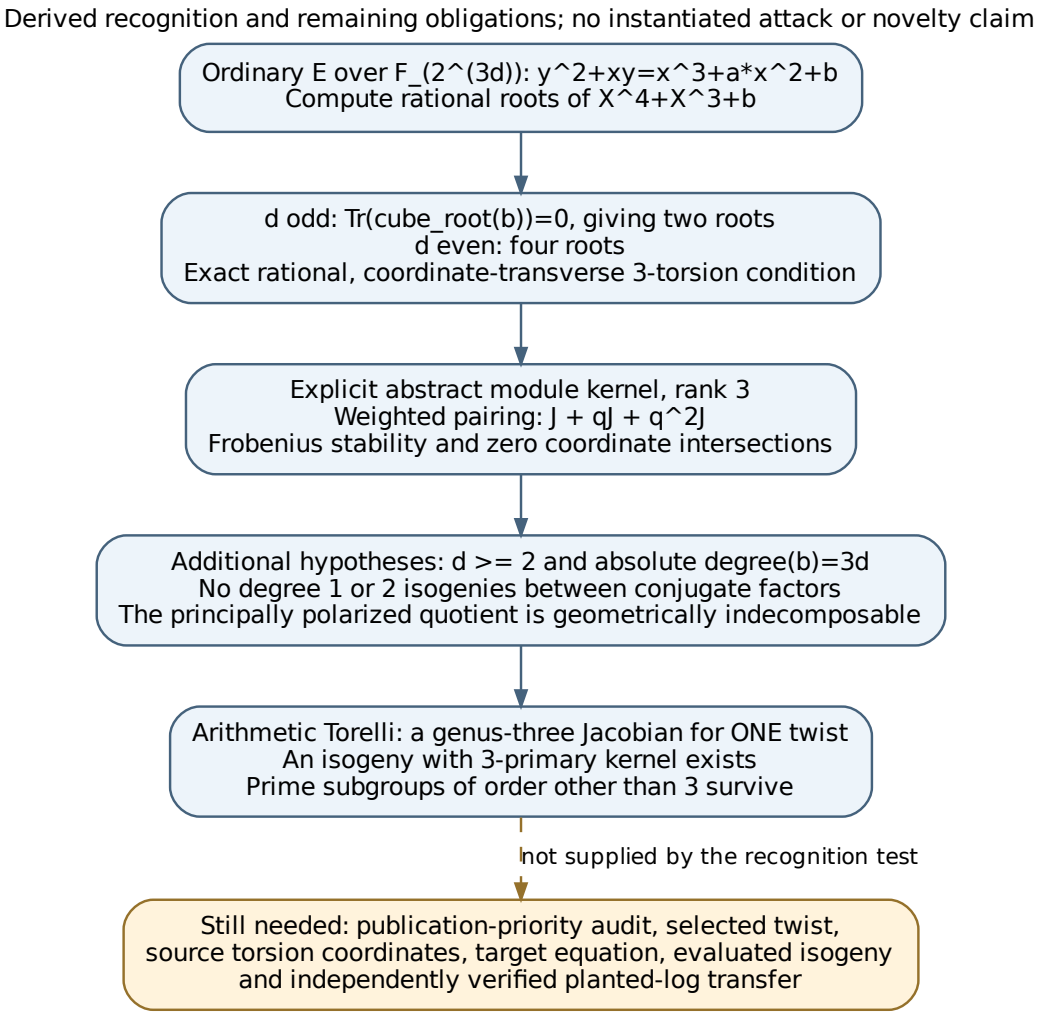


Figure 1: Recognition, geometric and explicit-transfer obligations are separate.

Native Rust boundary and next search

`examples/three_gluing.rs` checks a supplied mod-three matrix, constructs the explicit module kernel in the positive cases, and verifies its pairing, stability and intersections. Its binary-coefficient mode applies the polynomial-gcd test, checks full absolute degree and preserves all remaining twist, target and explicit-map obligations. It does not change the frozen sweeper engine or existing journal fingerprints.

```
cargo run --release --locked \
  --manifest-path experiments/ghs-sweeper/Cargo.toml \
  --example three_gluing -- curve --base-degree 3 --a 0 --b 24
cargo run --release --locked \
  --manifest-path experiments/ghs-sweeper/Cargo.toml \
  --example three_gluing -- matrix --q-mod-3 2 --entries 1 0 0 2
```

The independent unit code checks all 48 invertible two-by-two matrices against a direct enumeration of the 1120 Lagrangians for each weighted form. Fixed coefficient controls compare polynomial-gcd counts with direct raw-field evaluation. See THREE_GLUING_CHECKS.md. These are software checks of a fixed algebraic statement, not new curve-search campaign data or a completed cryptanalytic experiment.

Fixed software observations: all 29 release all-target tests pass, including the four new tests. The direct subspace check agrees with the criterion on all 48 invertible matrices, using 1120 Lagrangians per weighted form. Formatting and all-target Clippy with warnings denied pass. At the fixed degree-nine coefficient $b = 24$ (modulus 515), both supplied constants $a = 0, 1$ give two rational division roots, and b has absolute degree nine. This verifies the recognition condition for that twist pair; it does not select a twist or instantiate its target. The raw unit log retains the final software result; the earlier 28-test output is retained too. The closed trace test and its two explicit division roots agree with the polynomial-gcd result on the fixed odd-degree controls. No curve-population or cost comparison is made, so no quantitative curve-yield or performance graph is inferred from these checks.

The next mathematical/computational obligations are: priority review of this exact criterion; binary quotient reconstruction; the characteristic-two obstruction character; source-coordinate torsion certificates; explicit isogeny evaluation and planted-log replay. Only then can an exact curve and subgroup enter the weak-curve catalogue. No cost comparison is required to identify an affected family, but no measured cost is asserted here.

The larger campaign's 355 jobs remain pending: the required protocol tool is absent from inspected `crypto-autoresearcher origin/main` at `4ea4e6a22ab30e4da6f4b7515e4fe2dfbdfa8a58`. No substitute protocol was used. This leaves the proof and software work above available.

Evidence, source gaps and graph audit

- **Read primary:** Tian, 2020 preprint, sections 1 and 3; establishes the isogeny-based cover-attack prior art and necessary coordinate condition. arXiv:2012.07173.
- **Read primary:** Beauville–Ritzenthaler, introduction and section 3, especially Proposition 3.1 and the characteristic-two discussion. arXiv:1005.3683.
- **Read primary:** Lachaud–Ritzenthaler–Zykin, Theorem 1.1.1, gives the any-field arithmetic Torelli alternative. arXiv:0802.4017.
- **Read primary:** Moisio, section 7, Theorem 7.3 and Remark 7.4, documents the older trace-of-cube-root condition in the binary Kloosterman setting. arXiv:0706.2112.
- **Prior local work:** Hess baseline HESS_SCOPE.md, old prior-art comparison, and Diem–Scholten's 2003 cover-attack report. These do not establish priority for this formulation.
- **Unread:** the full 2023 Tian journal version and restricted Murai thesis. Neither is represented as ruled out. Literature searches for binary cubic gluing, scalar Frobenius and rational 3-torsion did not establish novelty.

Claims above are derived mathematical statements, not independently reviewed mathematical results or Lean-checked theorems. Exact source-curve and subgroup IDs, candidate IDs and run IDs for attacks are null: no instantiated transfer is supplied. Measured construction, transfer, solve and rho costs are null.

Graph audit: checked the infrastructure source-parity table, GHS and branch-orbit reports, curve registry and route index. No concrete curve or verified route was added; existing graphs remain current. The new diagram records theorem implications and remaining obligations.